Series 4

Even though this PVK is structured into thematic sections and some content has not yet been covered, it is explicitly permitted and *should* be the case that in the problem sets *all the material* covered in the semester and in the lecture may be used.

Exercise 1 (FS2024 A.Figalli)
Let $f : [0, \infty) \to \mathbb{R}$ be a continuous function such that $f(0) = 0$ and the limit $L = \lim_{x\to\infty} f(x)$ exists.

(a) Show that f is bounded.

(b) Let m be the infimum of f on $\mathbb{R}$. Show that $m \leq 0$.

(c) Suppose that $L \geq 0$, show that f attains its minimum.

(d) Show that f does not necessarily have to attain its minimum if $L < 0$.

Solution:

(a) Since we know that the limit L exists, by definition for every arbitrary ε there exists an R such that

$$|f(x) - L| < \varepsilon \quad \forall x \geq R.$$

Choose $\varepsilon = 1$. Then there exists an $R \in \mathbb{R}$, such that

$$|f(x)| = |f(x) - L + L| \leq |f(x) - L| + |L| < 1 + |L| \quad \forall x \geq R.$$

Thus $f(x)$ is bounded on $[R, \infty)$. Since f is continuous and $[0, R]$ is compact, f is bounded on all of $[0, \infty)$.

(b) Since the infimum is the greatest lower bound, we have that the infimum $m \leq x$ for all $x \in \mathrm{Im}(f)$. Because we know that $f(0) = 0$, 0 is in the image of f, so it follows for the infimum m that

$$m \leq 0.$$

(c) Let m be the minimum of f. We already know that $m \leq 0$. If $m = 0$, then f attains this at $x = 0$.
Now assume $m < 0$. Choose $\varepsilon = |m|$, then from the existence of the limit we have

$$|f(x) - L| < |m| \quad \forall x \geq N \in \mathbb{R}.$$

Since $L \geq 0$ we have

$$f(x) > -|m| \quad \forall x \geq N.$$

Because $[0, N]$ is compact and f is continuous, f attains its minimum there.

(d) A simple counterexample is the function $f(x) = e^{-x} - 1$. This function never attains its minimum $m = -1$.

Exercise 2
Solve the initial value problem:

$$\begin{cases} y'(x^2 + 1) - 2xy = 0, \\ y(0) = 1. \end{cases}$$

Solution: We see that this differential equation is separable. Therefore we have

$$\frac{dy}{dx} = \frac{2x}{x^2+1}y \iff \int \frac{1}{y}dy = \int \frac{2x}{x^2+1}dx.$$

We solve the second integral via substitution $u = x^2$, $du = 2x\,dx$. Thus

$$\log(y) = \log(x^2+1) + C \iff y = A(x^2+1).$$

From the boundary condition we find $A = 1$, hence the solution is

$$y = x^2 + 1.$$

Exercise 3 (FS2024, A. Figalli)
Solve the initial value problem:

$$\begin{cases} u'' + u' - 2u = -2x, \\ u(0) = \frac{3}{2}, \\ u'(0) = 2. \end{cases}$$

Solution: First we look for a homogeneous solution. For this we set up the characteristic polynomial

$$\lambda^2 + \lambda - 2 = 0 \iff (\lambda - 1)(\lambda + 2) = 0.$$

For our homogeneous ansatz we thus have

$$u_{\text{Hom.}} = Ae^x + Be^{-2x}.$$

As a particular ansatz we take a polynomial of first degree $u_{\text{Part.}} = Cx + D$.

$$C - 2Cx - 2D = -2x \iff C = 1, \quad D = \frac{1}{2}$$

From the boundary conditions we find $A = 1$ and $B = 0$. Thus the solution is

$$u(x) = e^x + x + \frac{1}{2}.$$

Exercise 4 (FS2024, A. Figalli)
Compute the integral

$$\int_0^1 \frac{1}{\sqrt{x(1-x)}}dx.$$

Solution: We use the substitution $x = \sin^2(u)$, from which it follows that $\frac{dx}{du} = 2\cos(u)\sin(u)$ and

$$\int_0^1 \frac{1}{\sqrt{x(1-x)}}dx = \int_0^{\frac{\pi}{2}} \frac{2\cos(u)\sin(u)}{\sqrt{\sin^2(u)(1-\sin^2(u))}}du = \int_0^{\frac{\pi}{2}} \frac{2\cos(u)\sin(u)}{\sin(u)\cos(u)}du = \pi.$$

Exercise 5 (FS2024, A. Figalli)
Compute the integral

$$\int \frac{2x+1}{x(x-1)}dx.$$

Solution: We know via partial fraction decomposition

$$\int \frac{2x+1}{x(x-1)}dx = \int \frac{3}{x-1} - \frac{1}{x}dx = 3\log|x-1| - \log|x| + C.$$

Exercise 6 (FS2024, A. Figalli)
Compute the integral

$$\int \frac{\sin(x)}{\cos(x)+1}dx.$$

Solution: We use the substitution $u = \cos(x)$, from which it follows that $\frac{du}{dx} = -\sin(x)$. Therefore

$$\int \frac{\sin(x)}{\cos(x)+1}dx = -\int \frac{1}{u+1}du = -\log|\cos(x)+1| + C.$$

Exercise 7 (HS2023, A. Figalli)
Compute the Taylor polynomial at 0 up to third degree of the function

$$\cos^2(x) - \cos(x^2)e^{x^2}.$$

Use Big-O Landau notation to indicate the remainder term.

Solution: We insert the first terms of the Taylor expansion of the exponential function and the cosine. Set $t = x^2$ and note that already $t^2 = x^4 = O(x^3)$.

$$\left(1 - \frac{x^2}{2} + O(x^3)\right)^2 - \left(1 + O(t^2)\right)\left(1 + t + O(t^3)\right) =$$

$$1 - x^2 + O(x^3) - \left(1 + t + O(t^2)\right) = 1 - x^2 - 1 - x^2 + O(x^3) = -2x^2 + O(x^3).$$

Exercise 8 (FS2024, A. Figalli)
Compute the Taylor polynomial at 0 up to fourth degree of the function

$$\sin(x^2) - \sin^2(x).$$

Use Big-O Landau notation to indicate the remainder term.

Solution: We insert the first terms of the Taylor expansion of the sine. Set $t = x^2$,

$$t + O(t^3) - \left(x - \frac{x^3}{6} + O(x^3)\right)^2 = x^2 - x^2 + \frac{x^4}{3} + O(x^4) = \frac{x^4}{3} + O(x^4).$$

Exercise 9
Compute the integral

$$\int \log(1 + x^2) dx.$$

Solution:

$$\int \log(1 + x^2) dx = x \log(1 + x^2) - \int \frac{2x^2}{1 + x^2} dx = x \log(1 + x^2) - 2 \int 1 - \frac{1}{1 + x^2} dx$$

$$= x \log(1 + x^2) - 2x + 2 \arctan(x) + C.$$

Tip: Further integrals for practice can be found in the Moodle integral trainer and in the exam collection. Further ODEs can be found in series 13 (HS2025).